Lecture 19: Internet E-Mail Format Protocol (*RFC 822*) & Simple Mail Transfer Protocol, SMTP (*RFC 821*) - I

Objectives:

- Learn the Format of an Internet E-Mail
- Learn the basic components of E-Mail system
- Learn the basic SMTP commands and reply codes

1. Mail Format Protocol

The RFC 822 describes the structure and content of TCP/IP e-mail messages. The structure is intentionally designed to be very simple and easy to both create and understand.

Each message begins with a set of headers that describe the message and its contents. An empty line marks the end of the headers, and then the message body.

The headers help control how the message is processed, by specifying who the recipients are, describing the contents of the message, and providing information about processing done on the message as it was delivered.

1.1 Header Format

Each header field follows the simple text structure we saw in http protocol:

<header name>: <header value>

Like all RFC 822 lines, headers must be no more than 998 characters long and are recommended to be no more than 78 characters in length, for easier readability.

The RFC 822 supports a special syntax for allowing headers to be "folded" onto multiple lines if they are very lengthy. This is done by continuing a header value onto a new line, which *must* begin with at least one "white space" character, such as a space or <Tab> character, like this:

<header name>: <header value part 1>
<white space> <header value part 2>
<white space> <header value part 3>
…

The <Tab> character is most often used for this purpose. So, for example, if we wanted to specify a large number of recipients for a message, we could do it as follows:

To: person1@domain1.org, person2@domain2.com
person3@domain3.net, person4@domain4.edu

1.2 Header Field Groups

To help organize the many headers, they are categorized into header field groups (as did RFC 822,

- **Origination Date Field:** Specifies the date and time that the message was made ready for delivery; see below for details.
- **Originator Fields:** Contain information about the sender of the message.
- **Destination Address Fields:** Specify the recipient(s) of the message, which may be in one of three different recipient classes.
- **Identification Fields:** Contain information to help identify the message.
- **Informational Fields:** Contain optional information to help make more clear to the recipient what the message is about.
- **Resent Fields:** Used to preserve the original originator, destination and other fields when a message is resent.
- **Trace Fields:** Used to show the path taken by mail as it was transported.

In addition, the format allows other, user-defined fields to be specified, as long as they correspond to the standard "<header name>: <header value>" syntax.

1.3 ***Common Header Field Groups and Header Fields***

This table describes the header fields and how they are used:

Field Group	Field Name	Appearance	Description
Origination Date	***Date:***	Mandatory	Indicates the date and time that the message was made available for delivery by the mail transport system.
Originator Fields	***From:***	Mandatory	The e-mail address of the user sending the message. This should be the person who was the source of the message itself.
	Sender:	Optional	The e-mail address of the person who is sending the electronic mail, if different from the message originator. For example, if person *B* is sending an e-mail containing a message from person *A* on *A*'s behalf.
	Reply-To:	Optional	Used to tell the recipient of this message the address the originator would like the recipient to use for replies. If absent, replies are normally sent back to the *From:* address.
Destination Address Field	***To:***	Normally present	A list of primary recipients of the message.
	Cc:	Optional	A list of recipients to receive a "copy" of the message.

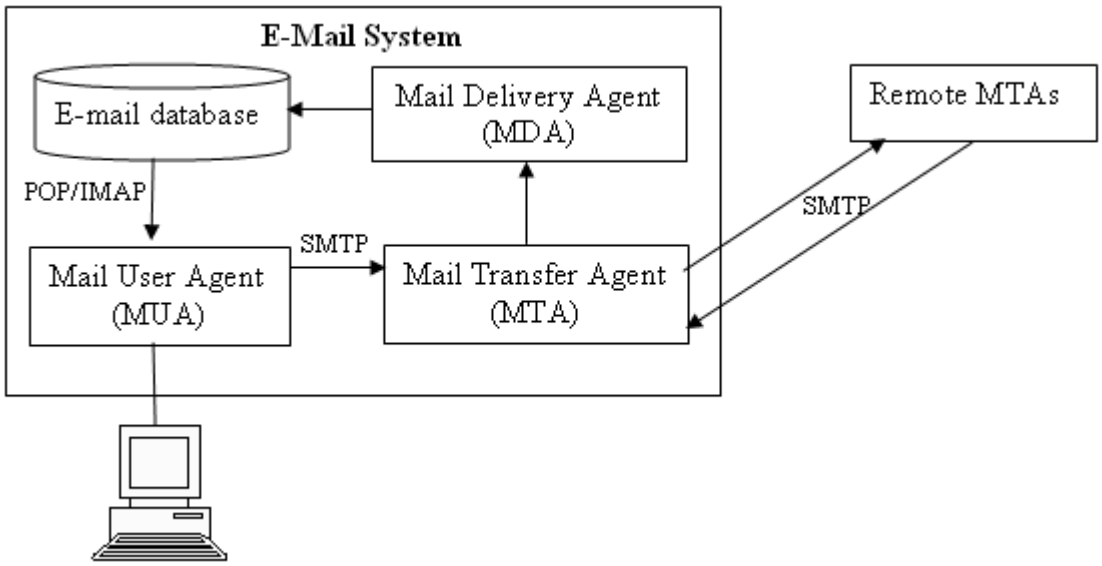
	Bcc:	Optional	Contains a list of recipients to receive a "blind carbon copy". These people receive a copy of the message without other recipients knowing they have received it.
Identification Fields	***Message-ID:***	Should be present	Provides a unique code for identifying a message; normally generated when a message is sent.
	In-Reply-To:	Optional, normally present for replies	When a message is sent in reply to another, the *Message-ID:* field of the original message is specified in this field.
	References:	Optional	Identifies other documents related to this message, such as other e-mail messages.
Informational Fields	***Subject:***	Normally present	Describes the subject or topic of the message.
	Comments:	Optional	Contains summarized comments about the message.
	Keywords:	Optional	Contains a list of comma-separated keywords that may be of use to the recipient.
Resent Fields	***Resent-Date: Resent-From: Resent-Sender: Resent-To: Resent-Cc: Resent-Bcc: Resent-Message-ID:***	Each time a message is resent, a resent block is required	These are special fields used only when a message is re-sent by the original recipient to someone else, a process called *forwarding*.
Trace Fields	***Received: Return-***	Inserted by e-mail system	These fields are inserted by computers as they process a

	Path:		message and transport it from the originator to the recipient.

2. E-Mail Basics

Most e-mail systems are based on the UNIX e-mail model. In this model, e-mail functions and divided into three software modules, namely:

- Mail User Agent (MUA)
- Mail Transfer Agent (MTA)
- Mail Delivery Agent (MDA)



2.1 Mail User Agent (MUA):

These are applications that allow users to access their mails residing on a mail server either by downloading them to the local machine (using **POP**) or by viewing them directly on the server (using **IMAP**).

Most MUAs also allow users to compose mails. The mails are forwarded to the MTA (using SMTP) for onward delivery. Note that MUAs do not deliver messages to their target by themselves.

Common MUAs are Microsoft Outlook and Pine.

2.2 Mail Transfer Agent (MTA):

MTA applications handle both incoming and outgoing messages using SMTP.

For each outgoing message, the MTA determines the destination of the recipient address.

If the destination is local, the MTA can either deliver it directly to the local mailbox, or pass it to the MDA for delivery.

If the destination is remote, the MTA establishes connection with the remote MTA using SMTP to deliver the message.

A good example of MTA is **SendMail** (a utility in UNIX). IIS also has an SMTP server that acts both as an MTA and as an MDA.

2.3 Message Delivery Agent (MDA):

The main purpose of MDA application is to deliver mails to user accounts. Thus, MDAs must know the location of user mail-boxes.

Example of MDA is the **procmail** (a utility in UNIX). In IIS, the functionalities of MTA and MDA are provided by a component called virtual SMTP server.

3. Overview of SMTP

The purpose of the Simple Mail Transfer Protocol (SMTP) is to provide rules to follow in sending and receiving of mails between two communicating partners in a reliable manner.

An SMTP server is normally implemented using TCP and it uses the well-known port 25.

Once a client connects to an SMTP server, the server responds by sending a welcome message. The client can then communicate with the server using one of the following standard commands:

3.1 Basic SMTP Commands

HELO *sendinghostname*	Initiates SMTP conversation. The host connecting to the remote SMTP server identifies itself by it's fully qualified host name.
MAIL From:<*source email address*>	This is the start of an email message. The source email address is what will appear in the "From:" field of the message.
RCPT To:<destination email address>	This identifies the recipient of the email message. This command can be repeated multiple times for a given message in order to deliver a single message to multiple recipients.
DATA	This indicates the start of the email message body. The stream of data is terminated by a "." on a line by itself.
QUIT	This terminates an SMTP connection. Multiple email messages can be transferred during a single connection. To start another email message in the same session, simply issue another "MAIL" command.
Other commands are: VRFY, EXPN, NOOP, RSET, HELP, etc.	

3.2 SMTP Extension (RFC 2554)

Due to the problem of mail spam, SMTP has been extended to provide authentication. This is done by adding the **AUTH LOGIN** command.

This is used to supply username and password (in Base64 encoding).

Both the ITC mail server (smtp.kfupm.edu.sa) and CCSE mail server (mailer.ccse.kfupm.edu.sa) require authentication.

3.3 Reply Codes:

For each Command sent to an SMTP server, the sever sends back a single reply consisting of a 3 digit code and a text comment.

The first digit denotes whether the response is good, bad or

incomplete. An unsophisticated sender-SMTP will be able to determine its next action (proceed as planned, redo, retrench, etc.) by simply examining this first digit.

A sender-SMTP that wants to know approximately what kind of error occurred (e.g., mail system error, command syntax error) may examine the second digit.

The third digit provides the finest gradation of information.

There are five values for the first digit of the reply code:

1yz Positive Preliminary reply
The command has been accepted, but the requested action is being held pending confirmation of the information in this reply.

The sender-SMTP should send another command specifying whether to continue or abort the action.

[Note: SMTP does not have any commands that allow this type of reply, and so does not have the continue or abort commands.]

2yz Positive Completion reply
The requested action has been successfully completed. A new request may be initiated.

3yz Positive Intermediate reply
The command has been accepted, but the requested action is being held pending receipt of further information. The sender-SMTP should send another command specifying this information.

4yz Transient Negative Completion reply
The command was not accepted and the requested action did not occur. However, the error condition is temporary and the action may be requested again.

5yz Permanent Negative Completion reply
The command was not accepted and the requested action did not occur. The sender-SMTP is discouraged from repeating the exact request (in the same sequence).

Sample Error-Codes and their meaning:

211	System status, or system help reply
214	Help message. [Information on how to use the receiver or the meaning of a particular non-standard command; this reply is useful only to the human user]
220	Service ready
221	Service closing transmission channel
250	Requested mail action okay, completed
251	User not local; will forward to
354	Start mail input; end with .
421	Service not available, closing transmission channel.
450	Requested mail action not taken: mailbox unavailable. [E.g., mailbox busy]
451	Requested action aborted: local error in processing
452	Requested action not taken: insufficient system storage
500	Syntax error, command unrecognized. [This may include errors such as command line too long]
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command parameter not implemented
550	Requested action not taken: mailbox unavailable. [E.g., mailbox not found, no access]
551	User not local; please try
552	Requested mail action aborted: exceeded storage allocation
553	Requested action not taken: mailbox name not allowed. [E.g., mailbox syntax incorrect]
554	Transaction failed

Note: The reply text may be longer than a single line; This requires a special format so the sender-SMTP knows when it can stop reading the reply.

The format for multi-line replies requires that every line, except the last, begin with the reply code, followed immediately by a hyphen, "-" (minus), followed by text. The last line will begin with the reply code, followed immediately by <SP>, optionally some text, and <CRLF>.

For example:

123-First line
123-Second line
123-234 text beginning with numbers
123 The last line

3.4 Sample SMTP Session:

The following is a sample client-server conversation. The text in blue is the client input.

```
telnet smtp.kfupm.edu.sa 25
220 smtp.kfupm.edu.sa ESMTP ITC-KFUPM MAIL Server ready at Thu, 27 Nov 2003 17:2
2:13 +0300 - SPAM Prohibited
helo localhost
250 smtp.kfupm.edu.sa Hello ics-bmghandi.pc.ccse.kfupm.edu.sa [196.1.65.143], pleased to
meet you
auth login
334 VXNlcm5hbWU6
Z21iYXNoaXI=
334 UGFzc3dvcmQ6
c3dlMzQ0MDMx
235 2.0.0 OK Authenticated
MAIL FROM: gmbashir@kfupm.edu.sa
250 2.1.0 gmbashir@kfupm.edu.sa... Sender ok
RCPT TO: bmghandi@yahoo.co.uk
250 2.1.5 bmghandi@yahoo.co.uk... Recipient ok
DATA
354 Enter mail, end with "." on a line by itself
From: Bashir Ghandi <gmbashir@kfupm.edu.sa>
Subject: Testing SMTP Server at KFUPM

This is just to test the SMTP server at KFUPM.
Please ignore this mail.
Thanks,
Bashir.
.
250 2.0.0 hAREMDql018976 Message accepted for delivery
quit
221 2.0.0 smtp.kfupm.edu.sa closing connection


Connection to host lost.
```